

Introduction

My PhD project is part of a larger project called DP-Next. DP-Next is a project aimed at developing a sustainably effective strategy for prevention of Type 2 Diabetes in Denmark, Greenland and the Faroe Islands.

My goal is to develop a risk prediction model that can predict the 5 year risk of getting diagnosed with type 2 diabetes and the 5 year risk of death, only using registry data. This model is to be used to select people that will be included in the development of the prevention strategy.

Motivation/Setting

When you estimate the cause specific risks at time τ using semi-parametric Cox regression for the cause specific hazards, the sum of all cause specific risks can be more than 1 for certain covariate values.

We need to export the prediction model developed in protected environments, such that we can give it to the public.

It is unclear when to artificial censor subjects at the prediction time horizon.

Setting: competing risks

Assume we are in a competing risks scenario and want to predict the risk of getting diagnosed with T2D and the risk of dying, at some fixed timepoint τ . Denote:

- T as the time between time0 and the date of an event
- $D \in \{1, 2\}$ as the cause of the event
- $X = (X^1, \dots, X^p)$ as a p -dimensional vector of baseline covariates, which can both be numeric or categorical.

Additionally we assume that the event time T is right censored at the random time C , where $P(C > \tau | X) > 0$ almost surely. Denote $\tilde{T} = \min(T, C)$ and $\tilde{D} = \Delta D$ as the observed time and event, where $\Delta = 1\{T \leq C\}$.

Setting: Proportional hazards model

Let β be a p -dimensional parameter vector and x be some observed value of X . We are interested in the cumulative incidence function for both causes, which for cause j is $F_j(t|x) = P(T \leq t, D = j|x)$, and can be written as

$$F_j(t|x) = \int_0^t S(s-|x) d\Lambda_j(s|x),$$

where S is the survival function

$$S(t|x) = \exp \left(- \sum_{i=1}^2 \Lambda_i(s|x) \right),$$

and Λ_j is the cumulative hazard

$$\Lambda_j(t|x) = \exp(x\beta_j) \Lambda_{0j}(t).$$

Λ_{0j} is the baseline cumulative hazard for cause j , which is defined as $\Lambda_{0j}(t) = \int_0^t \lambda_{0j}(s) ds$, where λ_{0j} is the baseline hazard

Proportional hazards model estimation

The typical ways of estimating these kinds of models is to either use partial likelihood, where we assume the baseline hazard is non-parametric, or maximum likelihood, where we need to choose some parametric form for the baseline hazard functions.

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I will throughout the presentation assume that C and T are conditionally independent given X .

Maximum likelihood

Assume we have N observations of $(\tilde{T}, \tilde{D}, X)$. Using that the survival function is a product of the individual cause-specific survival functions, $S(t_i|x_i) = \prod S_j(t_i|x_i)$, we can write the likelihood as

$$L = \prod_{j=1}^2 \prod_{i=1}^N \lambda_j(t_i|x_i)^{d_{ij}} \exp(-\Lambda_j(t_i|x_i)),$$

where $d_{ij} = 1\{\tilde{D}_i = j\}$.

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where $d_{ij} = 1\{\tilde{D}_i = j\}$. This means that the likelihood function is a product of 2 likelihoods, one for each cause. We can then maximize each cause-specific likelihood function separately by choosing some parametric form for the baseline hazards.

Brier score estimation

The model I will be focusing on is the Cox-exponential model. This model assumes that the baseline hazard is constant, meaning that

$$\Lambda_{0j}(\tau) = \int_0^\tau \lambda_{0j}(s) ds = \tau \lambda_{0j},$$

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I will then estimate the parameters by minimizing the Brier-score for some fixed timepoint τ , i.e.

$$BSE = \arg \min_{\beta \in \mathbb{R}^{3 \times p}, \lambda_0 \in \mathbb{R}^{+3}} \sum_{j=0}^2 \frac{1}{N} \sum_{i=1}^N (F_j(\tau | x_i, \beta, \lambda_0) - o_i(\tau, j))^2,$$

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Assuming constant baseline hazards for all causes and letting

$A_i = \sum_{k=0}^2 \exp(\beta_k x_i) \lambda_{0k}$, we can write the BSE as

$$BSE = \arg \min_{\beta \in \mathbb{R}^{3 \times p}, \lambda_0 \in \mathbb{R}^{+3}} \sum_{j=0}^2 \sum_{i=1}^N \left(\exp(\beta_j x_i) \lambda_{0j} \frac{1 - \exp[-\tau A_i]}{A_i} - o_i(\tau, j) \right)^2$$

Artificial censoring

Assume we will artificially censor at $t = 5$. This means our new observed time will now be $\tilde{T}_{\text{trunc}} = \min(\tilde{T}, 5) = \min(T, C, 5)$ and the new observed event

$\tilde{D}_{\text{trunc}} = 1\{T \leq 5\} \tilde{D} = 1\{T \leq C \wedge T \leq 5\} D$. Doing artificial censoring means we truncate all times at a specific timepoint, and practically means that estimation methods such as the partial likelihood and maximum likelihood can't learn from data after that timepoint. The Brier-score estimation is unaffected by artificial censoring, as it only learns from data before timepoint τ .

Artificial censoring example plot using non-proportional hazards data

Practical examples

Using DP-Next data estimate for specific covariate values and fit models both with and without artificial censoring

Discussion

The BSE is not convex so it needs good initial guess to perform well.